

Acceso al sistema operativo y habilitación del SSH

#Resumen

Pasos para acceder al sistema operativo (Ubuntu) del carro y habilitar SSH para acceso remoto.

#Instrucciones

1. Conexión física y encendido

- Conecta el carro a una pantalla mediante HDMI y conecta un mouse al puerto USB.
- Enciende el carro (corriente directa o desde una power bank).



Figure 1: Car connected

2. Inicio de sesión

- El sistema operativo es Ubuntu. Credenciales:
 - Usuario: `deepracer`
 - Contraseña: `Steambog1$`

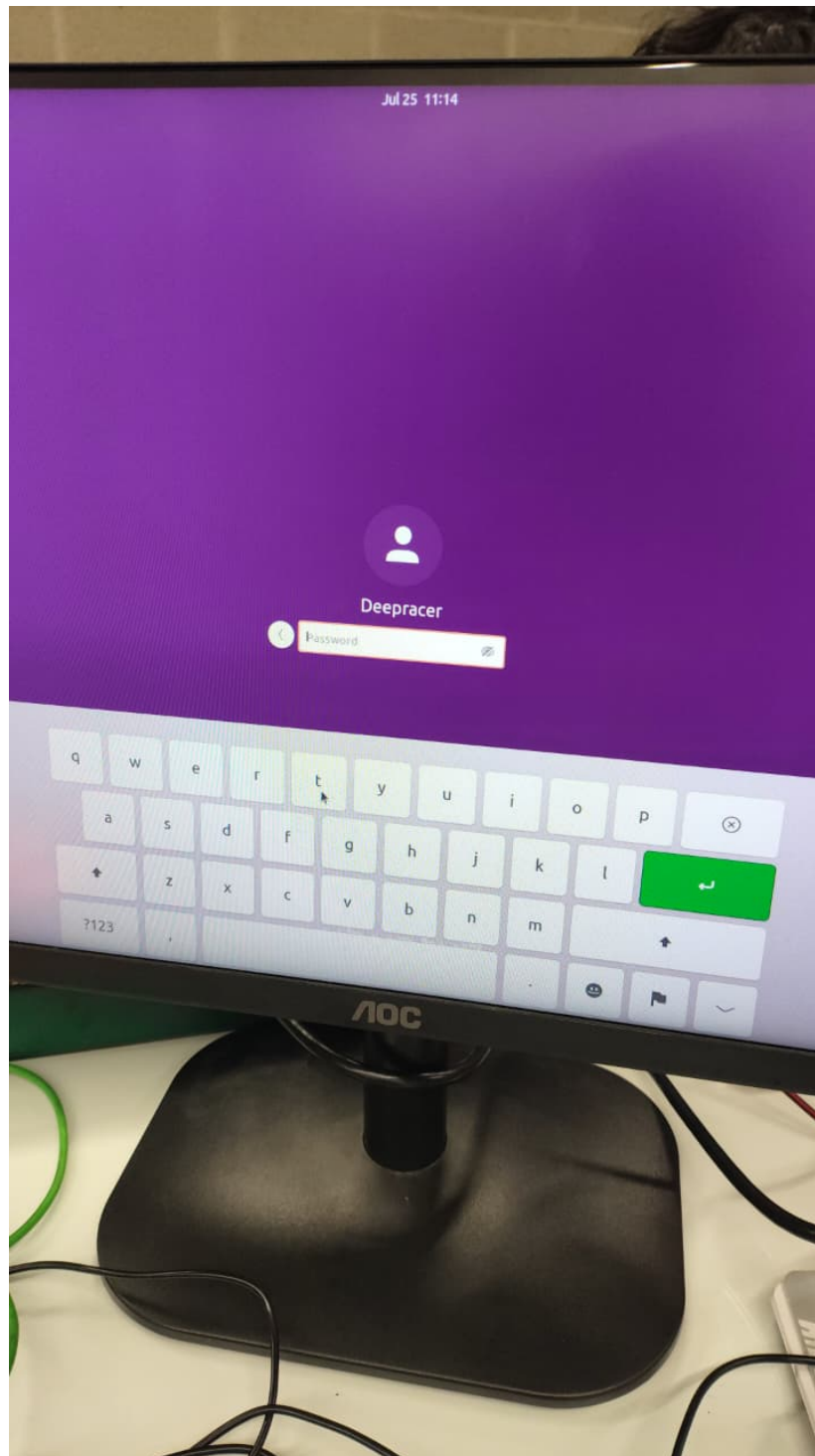
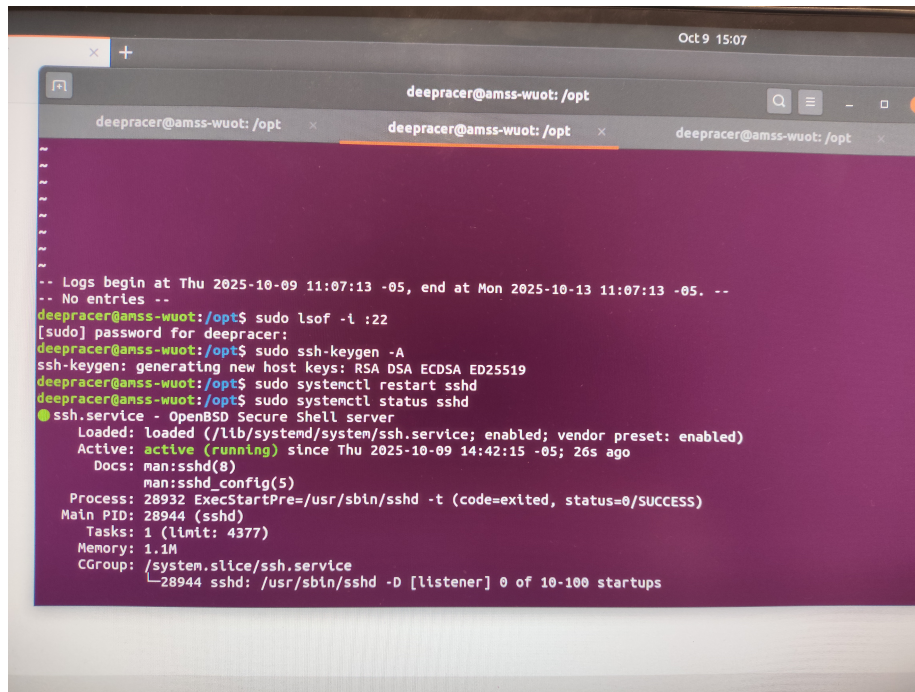


Figure 2: Pantalla de login

3. Sistema operativo (captura)

A screenshot of a terminal window with a dark purple background. The terminal shows a series of commands and their outputs. The user 'deepracer' is at the prompt 'deepracer@amss-wuot: /opt'. The commands executed are: 'sudo lsof -i :22' (output: '[sudo] password for deepracer:'), 'sudo ssh-keygen -A' (output: 'ssh-keygen: generating new host keys: RSA DSA ECDSA ED25519'), 'sudo systemctl restart sshd', and 'sudo systemctl status sshd'. The final output shows the status of the 'ssh.service' as 'active (running)' since 'Thu 2025-10-09 14:42:15 -05; 26s ago'. It also lists the process details for 'sshd' and the 'system.slice/ssh.service' group.

```
Oct 9 15:07
deepracer@amss-wuot: /opt
deepracer@amss-wuot: /opt
deepracer@amss-wuot: /opt
-- Logs begin at Thu 2025-10-09 11:07:13 -05, end at Mon 2025-10-13 11:07:13 -05. --
-- No entries --
deepracer@amss-wuot: /opt$ sudo lsof -i :22
[sudo] password for deepracer:
deepracer@amss-wuot: /opt$ sudo ssh-keygen -A
ssh-keygen: generating new host keys: RSA DSA ECDSA ED25519
deepracer@amss-wuot: /opt$ sudo systemctl restart sshd
deepracer@amss-wuot: /opt$ sudo systemctl status sshd
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2025-10-09 14:42:15 -05; 26s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 28932 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
    Main PID: 28944 (sshd)
       Tasks: 1 (limit: 4377)
      Memory: 1.1M
    CGroup: /system.slice/ssh.service
            └─28944 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
```

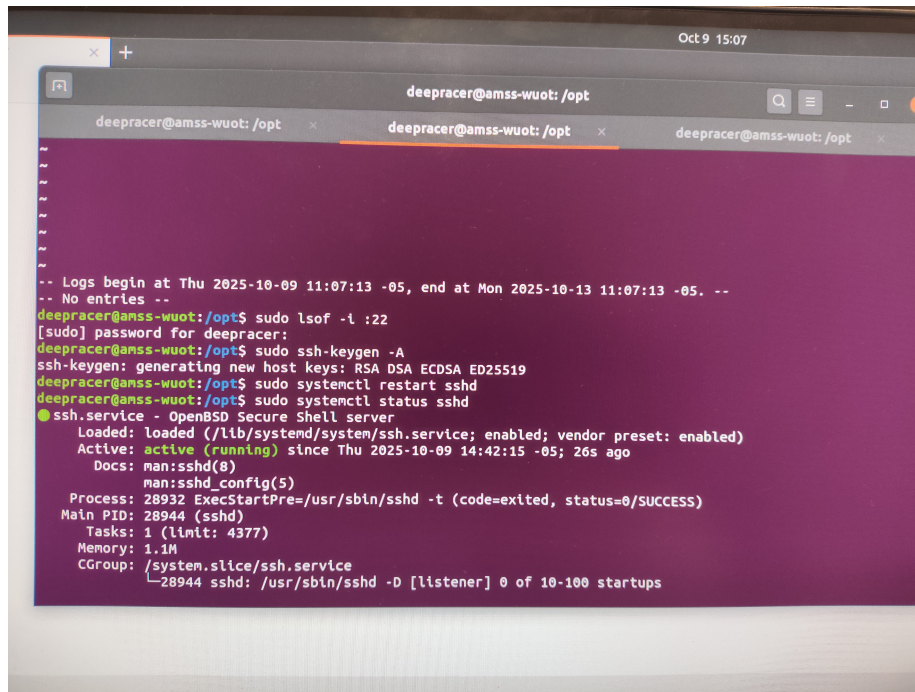
Figure 3: Operating system

4. Habilitar SSH

- Si el SSH no está habilitado y no se puede habilitar mediante la consola del AWS ni la terminal de Ubuntu, generar las claves del servidor ejecutando:

```
sudo ssh-keygen -A
```

Esto crea las claves host (si no existen) para los distintos algoritmos (rsa, ecdsa, ed25519, etc.). Estas claves las usa el servidor SSH para identificarse ante los clientes.



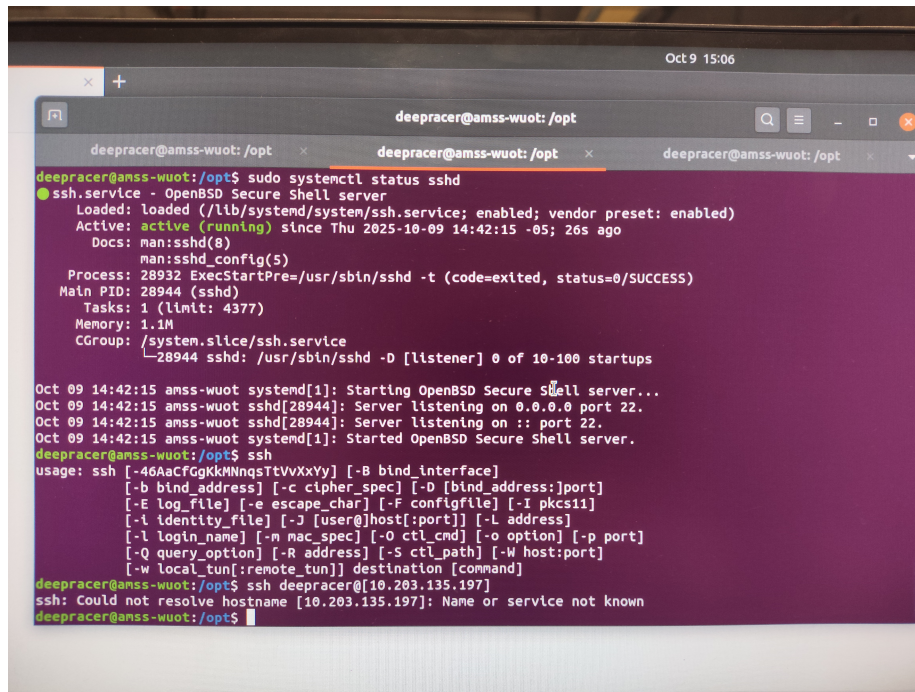
```
Oct 9 15:07
deepracer@amss-wuot: /opt
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deepracer@amss-wuot: /opt
-- Logs begin at Thu 2025-10-09 11:07:13 -05, end at Mon 2025-10-13 11:07:13 -05. --
-- No entries --
deepracer@amss-wuot: /opt$ sudo lsof -i :22
[sudo] password for deepracer:
deepracer@amss-wuot: /opt$ sudo ssh-keygen -A
ssh-keygen: generating new host keys: RSA DSA ECDSA ED25519
deepracer@amss-wuot: /opt$ sudo systemctl restart sshd
deepracer@amss-wuot: /opt$ sudo systemctl status sshd
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
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     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 28932 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
    Main PID: 28944 (sshd)
      Tasks: 1 (limit: 4377)
     Memory: 1.1M
    CGroup: /system.slice/ssh.service
            └─28944 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
```

Figure 4: Enable SSH

5. Comprobar el estado del servicio SSH

- Para ver si el servicio está corriendo:

```
sudo systemctl status sshd
```

```
Oct 9 15:06
deeppracer@amss-wuot: /opt
deeppracer@amss-wuot: /opt$ sudo systemctl status sshd
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2025-10-09 14:42:15 -05; 26s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 28932 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
  Main PID: 28944 (sshd)
    Tasks: 1 (limit: 4377)
   Memory: 1.1M
   CGroup: /system.slice/ssh.service
           └─28944 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups

Oct 09 14:42:15 amss-wuot systemd[1]: Starting OpenBSD Secure Shell server...
Oct 09 14:42:15 amss-wuot sshd[28944]: Server listening on 0.0.0.0 port 22.
Oct 09 14:42:15 amss-wuot sshd[28944]: Server listening on :: port 22.
Oct 09 14:42:15 amss-wuot systemd[1]: Started OpenBSD Secure Shell server.
deeppracer@amss-wuot: /opt$ ssh
usage: ssh [-46AaCfGgKkMMNnQsTtVvXxYy] [-B bind_interface]
           [-b bind_address] [-c cipher_spec] [-D [bind_address:]port]
           [-E log_file] [-e escape_char] [-F configfile] [-I pkcs11]
           [-i identity_file] [-J [user@]host[:port]] [-L address]
           [-l login_name] [-m mac_spec] [-O ctl_cmd] [-o option] [-p port]
           [-Q query_option] [-R address] [-S ctl_path] [-W host:port]
           [-w local_tun[:remote_tun]] destination [command]
deeppracer@amss-wuot: /opt$ ssh deeppracer@[10.203.135.197]
ssh: Could not resolve hostname [10.203.135.197]: Name or service not known
deeppracer@amss-wuot: /opt$
```

Figure 5: SSH status

6. Acceso por SSH desde otro equipo

- Asegúrate de que el equipo desde el que te conectas está en la misma LAN que el deeppracer.
- En el equipo remoto debe estar disponible un cliente SSH (en Linux y macOS viene por defecto; en Windows puedes instalar openSSH o alguna otra herramienta).
- Con la IP del deeppracer, conéctate así:

```
ssh deeppracer@IP_DEL_DEEPRACER
```

```

C:\Users\UNAL>ssh deeppracer@10.203.135.197
ssh: Could not resolve hostname [10.203.135.197]: Host desconocido.

C:\Users\UNAL>ssh deeppracer@10.203.135.197
The authenticity of host '10.203.135.197 (10.203.135.197)' can't be established.
ED25519 key fingerprint is SHA256:q0bm610TxxnC/Bkfyhoh8X2q83L5Vzpr4uHw654s.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.203.135.197' (ED25519) to the list of known hosts.
deeppracer@10.203.135.197's password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 4.15.0-1005-debplens x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

204 updates can be installed immediately.
121 of these updates are security updates.
To see these additional updates run: apt list --upgradable

Your Hardware Enablement Stack (HWE) is supported until April 2025.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

deeppracer@amss-wuot:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos
deeppracer@amss-wuot:~$
```

Figure 6: SSH access

Notas

- Si no conoces la IP del deeppracer, puedes obtenerla con `ip addr` o `ifconfig` en el dispositivo.
- En Windows 10/11 puedes usar `ssh` desde PowerShell: `ssh user@ip`